Close performance. Different trade-offs.

Comparative model performance | 16 October 2026

Main finding
XGBoost, LightGBM and CatBoost achieved closely matched test discrimination. Random Forest remained behind. Native categorical handling did not show a clear AUC advantage under the study budgets.

The test split contained **46,127 applicants** and **3,724 positive cases** (about 8.07%). Fourteen finalized pipelines were evaluated: eight one-hot and six native. Model and feature choices were based on validation; test rankings below are descriptive.

Tuned test results

Tuned pipeline	AUC	AP	Log loss	Top-10% recall
XGBoost / One-hot	0.7636	0.2533	0.2442	34.83%
CatBoost / One-hot	0.7635	0.2554	0.2443	34.88%
LightGBM / One-hot	0.7635	0.2525	0.2443	34.64%
XGBoost / Native	0.7633	0.2535	0.2443	34.85%
CatBoost / Native	0.7632	0.2546	0.2444	34.59%
LightGBM / Native	0.7624	0.2498	0.2448	34.45%
Random Forest / One-hot	0.7506	0.2377	0.2508	33.14%

Leading boosting models have closely matched AUC

One-hot XGBoost had the highest observed AUC, **0.7636**. One-hot CatBoost and LightGBM were both about **0.7635**. CatBoost led AP at **0.2554**, while XGBoost had the lowest log loss at **0.2442**. These small observed differences do not establish a decisive ordering of algorithms.

Paired tests cover tuning and native-versus-one-hot changes; direct paired score-difference intervals between the top three algorithms were not calculated. Overlapping individual intervals alone do not establish equivalence or a formal significance result.

Tuning helped, but gains were modest

All seven tuned pipelines improved observed test AUC. Improvement in the optimization metric did not guarantee improvement in every secondary metric.

Pipeline	AUC gain	Paired 95% interval
CatBoost | Native	+0.0019	[+0.0008, +0.0031]
CatBoost | One-hot	+0.0029	[+0.0016, +0.0042]
LightGBM | Native	+0.0016	[-0.0003, +0.0032]
LightGBM | One-hot	+0.0017	[+0.0001, +0.0037]
Random Forest | One-hot	+0.0055	[+0.0030, +0.0081]
XGBoost | Native	+0.0036	[+0.0015, +0.0055]
XGBoost | One-hot	+0.0022	[+0.0008, +0.0037]

Random Forest had the largest AUC gain (+0.0055), but its tuned AUC of 0.7506 remained below every boosting baseline in the displayed results. Native XGBoost had the largest AP gain, approximately **+0.0066**.

Native LightGBM's AUC interval crossed zero; its tuning gain is less certain under this bootstrap. One-hot LightGBM improved AUC but its AP moved slightly down, about **-0.0002**.

Native handling did not clearly outperform one-hot

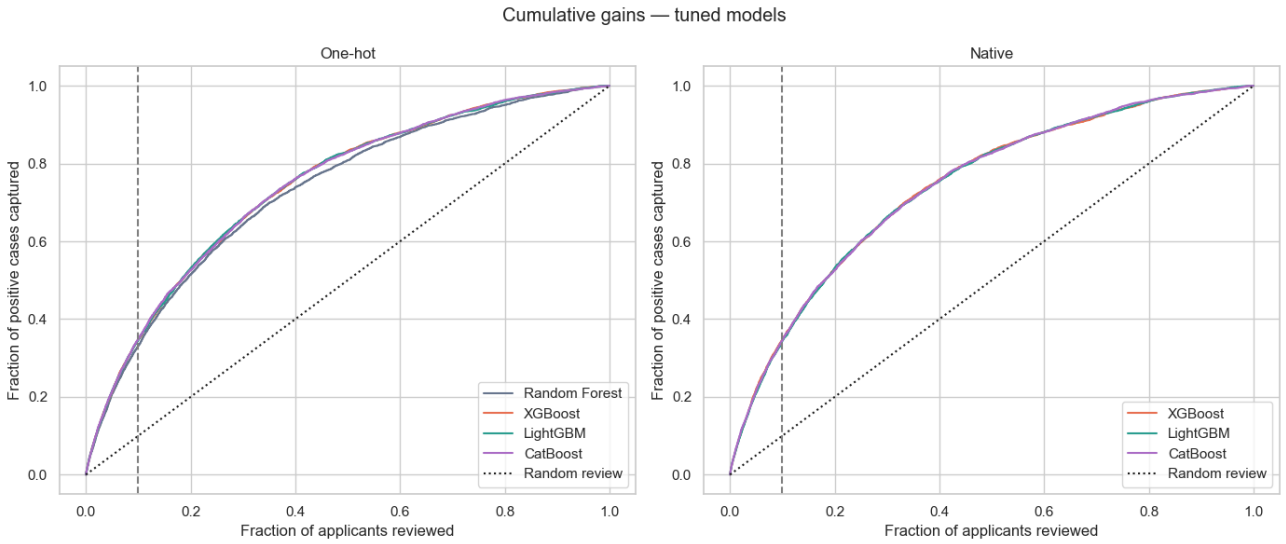
Algorithm	Native minus one-hot AUC	Paired 95% interval
CatBoost	-0.0003	[-0.0015, +0.0009]
LightGBM	-0.0011	[-0.0026, +0.0005]
XGBoost	-0.0003	[-0.0017, +0.0010]

All tuned native-versus-one-hot AUC intervals include zero. This supports **no clear observed native advantage**, not proof that the methods are identical. LightGBM and CatBoost selected different feature sets across representations, so their contrasts do not isolate categorical encoding alone.

Intervals are exploratory, pointwise bootstrap intervals for fixed models and this split. They do not capture training-seed variability or correct for multiple comparisons. Reported values are rounded from notebook outputs.

Useful prioritization, substantial missed cases

The strongest pipelines found about 35% of observed positive cases within the highest-risk 10% of applicants.



For one-hot CatBoost, top-10% recall was **34.88%**. Random review would capture about 10% in expectation, so this is roughly **3.5x concentration**. About 65% of positive cases still lay outside that review group. This is a retrospective ranking result, not prevented defaults or financial profit.

Errors at the frozen validation-F1 thresholds

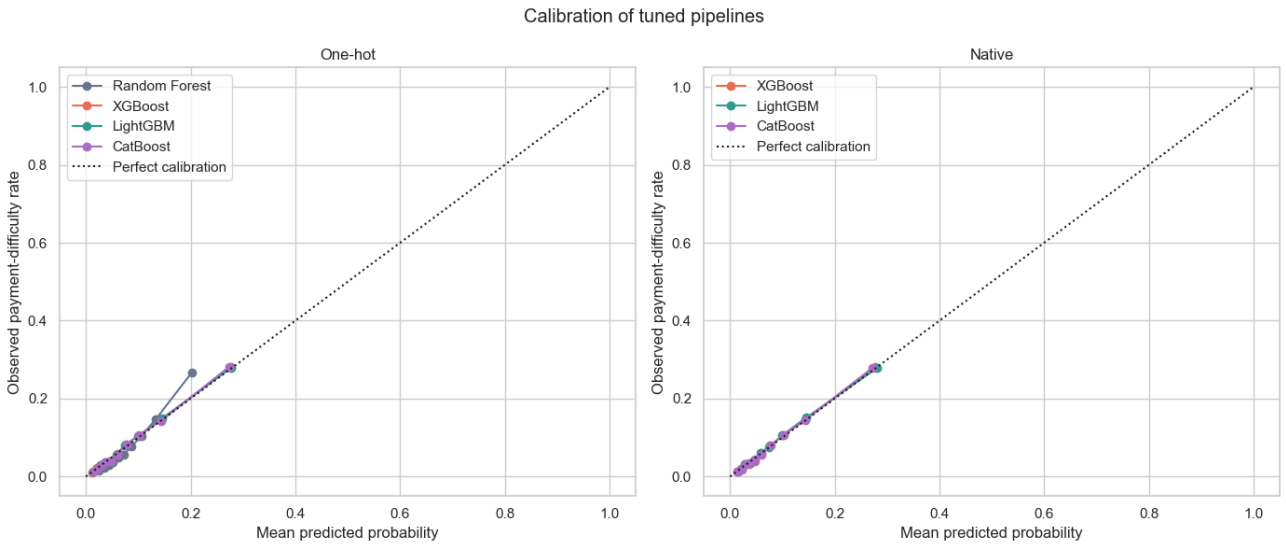
Validation-selected representative	One-hot CatBoost	Native XGBoost
Approximate threshold	0.147	0.147
Positive cases detected	1,632	1,616
Positive cases missed	2,092	2,108
False alerts	4,917	4,933
Test recall	43.82%	43.39%
Test precision	24.92%	24.68%

These representatives were chosen using validation AUC before test evaluation. At these thresholds, roughly **one in four flagged applicants** had the positive label. Different frozen thresholds create different alert volumes; the top-10% analysis provides a fixed-capacity comparison.

Operational interpretation
The models concentrate review effort, but many payment-difficulty cases are missed and most flags are false positives. A production threshold requires explicit costs, capacity and policy constraints.

Probabilities and predictive signals

Boosting calibration looked encouraging at the plotted bin averages; external scores were prominent predictors.

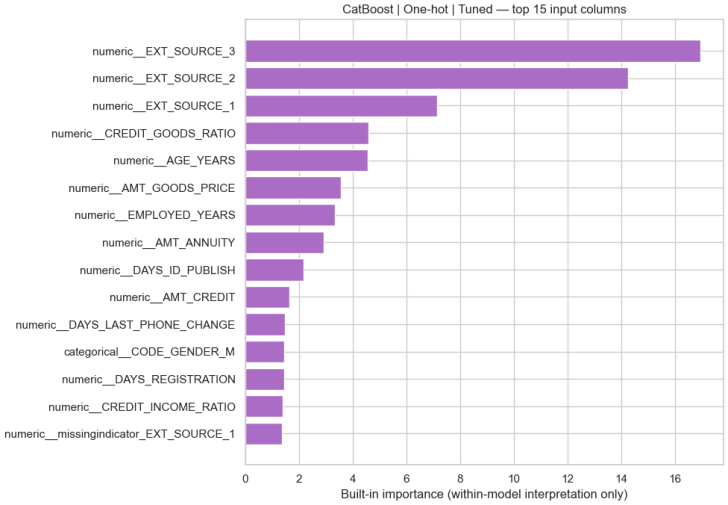


The boosting curves tracked the diagonal closely at the ten quantile-bin averages. Random Forest's highest-risk bin showed more underprediction: mean predicted risk near **0.20** corresponded to an observed rate near **0.26**. These are approximate visual readings, not precise bin-table estimates.

This supports reasonable agreement in the displayed bins, not perfect calibration for every applicant or subgroup. Log loss and Brier score reflect both discrimination and calibration. No calibration model was fitted after seeing the test outcomes.

Predictive signals in tuned one-hot CatBoost

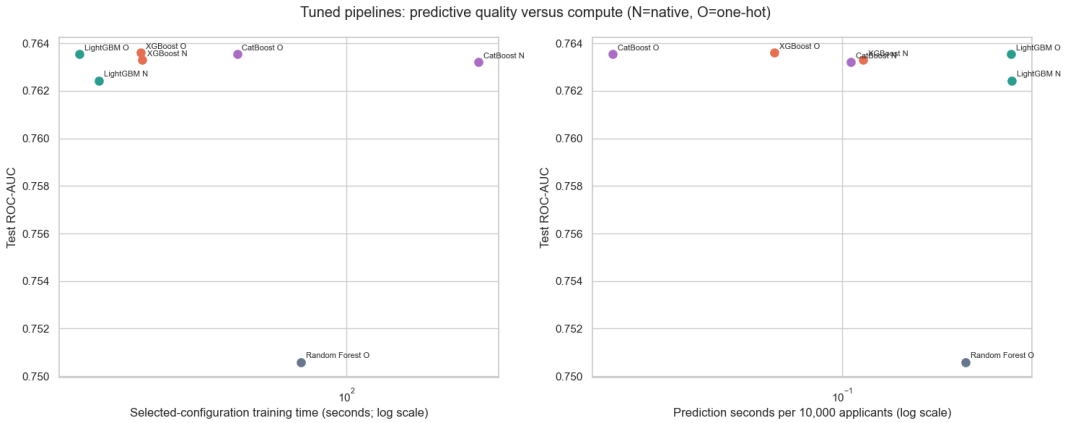
EXT_SOURCE_3, EXT_SOURCE_2 and EXT_SOURCE_1 were its three leading input columns. Credit-to-goods ratio, age, goods price and employment years were also prominent. This agrees with the earlier EDA observation that lower external-score groups showed more payment difficulty.



Built-in importance is within-model evidence, not causation. It also does not prove engineering improved results: that requires a comparison against a version without those engineered features.

Training and inference cost

There was no single speed winner for every task. Measurements describe the fitted configurations on the study machine.



Observed batch prediction times for tuned pipelines

Pipeline	Seconds / 10,000 validation applicants
CatBoost / one-hot	0.014
XGBoost / one-hot	0.056
CatBoost / native	0.108
XGBoost / native	0.119
Random Forest / one-hot	0.288
LightGBM / one-hot	0.426
LightGBM / native	0.429

One-hot LightGBM had the fastest selected-configuration fit in the plotted finalists. One-hot CatBoost had the fastest observed batch inference. Random Forest had by far the largest joblib file, approximately **160 MiB** from the plot; file size is not RAM usage.

Equal time allowed unequal exploration

Representation	RF	XGBoost	LightGBM	CatBoost
One-hot trials	4	58	73	38
Native trials	Not tested	70	68	7

Native CatBoost explored only seven trials; this is not its maximum possible performance. Most studies recorded about 30 minutes of trials; Random Forest recorded 33.2 minutes after overrun.

Prediction APIs can have different thread defaults; equal-thread inference speed superiority is not established without a thread audit. Loading and preprocessing were excluded.

Conclusion and study scope

Results describe the Home Credit dataset under the specified experimental protocol.

Study conclusion
On the Home Credit test split, tuned XGBoost, LightGBM and CatBoost achieved closely matched predictive performance and stronger observed scores than the Random Forest reference. Tuning delivered modest gains. Native categorical handling showed no clear AUC advantage over one-hot encoding under the allocated budgets. One-hot CatBoost combined strong average precision and fast measured batch prediction, while LightGBM offered fast fitting. The preferred pipeline depends on the operational trade-off rather than a decisive ranking winner.

Key findings

XGBoost one-hot had the highest observed test AUC. CatBoost one-hot had the highest observed AP and top-10% recall. Random Forest improved most in AUC from tuning but remained behind. The strongest pipelines captured about one-third of positives in a tenth of the applicants.

Interpretation boundaries

There is no direct paired top-three algorithm comparison in the notebook. No result proves native categoricals are inferior in general. Different selected features, different unseen-category treatment and limited trials can influence outcomes. Feature importance does not prove causality or the incremental benefit of engineering.

Study limitations

One stratified split and one training seed; repeated use of validation for development; exploratory bootstrap intervals; conclusions specific to this Home Credit protocol; suspected income-error row excluded without independent verification; fixed numeric imputation even for native models; potential inference-thread differences; no real lending cost, fairness or prospective deployment assessment.

Evidence and reproducibility

Source: recorded evaluation outputs and embedded figures in **04_home_credit_evaluation(1).ipynb**. No model was retrained for this brief. Metrics were read from the saved test and paired-comparison tables; approximate chart readings are labelled. Figures reproduce the recorded evaluation outputs.

Reproducibility materials include the frozen model registry, test prediction CSVs, paired-difference tables, thresholds, selected features, category definitions, trial history, package versions and hardware details.

Further robustness assessment would require direct paired comparisons among the top boosting pipelines and repeated training seeds, without using test outcomes for additional tuning.